Fast Pitch Detection Based on Improved FCM Algorithm and Hough Transform

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Abstract*—Football videos always appeals to large audiences in recent years. The pitch plays the fundamental role in automatically analyzing the factors in the game. In this paper, a fast pitch detection method based on an improved FCM algorithm and Hough transform is proposed. Firstly, use WFCM clustering algorithm for image processing. Secondly, determine the boundary by Hough transform, and then fill the region within the boundaries to get the mask. Hough transform is rarely used before, since it is difficult to detect the boundary from the many lines obtained by Hough transform. There are two reasons why we choose this method: the first one is that Hough transform is mature and fast;, secondly, a good segmentation result can be obtained after WFCM and corrosion processing which can reduce the number of the detected lines. Experimental results show that the proposed method is robust to various football videos.

Keywords-pitch detection; segmentation; WFCM; Hough transform;

1. Introduction

With the increasing popularity of sports video, more and more researchers have devoted themselves into this field. As we all know, the pitch detection is playing a pivotal role which contributes to video structure analysis as well as the follow-up detection of the players, the ball and the pitch markings. In addition, pitch detection also reduces the difficulty in three-dimensional animation reconstruction and improves the accuracy of video semantic analysis.

Now color information is widely used to detect pitch area, which is because the judgment of the main color will not be impacted as the colors of bitch are consistent. Jiang [1] and Wang [2] find the main color by classifying the quantized value of pixel according to the Gaussian mixture model. This method is effective except the model parameters can't be adjusted adaptively. Based on this, Liu et al. [3] get an improved result by proposing an adaptive Gaussian mixture model in pitch detection algorithm, but the shadow problems remains unsolved. In [4] the authors use pre-set green playfield space distribution area in the HSV space to detect the pitch, but this proposed method is restrictive. Ekin [5] proposes using the average value and the threshold value of the main color in any two color spaces of HSI, HSV, RGB and YCbCr spaces to detect the pitch pixels. The two color spaces will complement and control each other, inspired by which we optimize the sample by the characteristic values in two color spaces based on the color characteristic of the pitch.

In this paper, we use FCM algorithm to detect the pitch area at first which can achieve automatic segmentation even the characteristic of the image is uncertain and fuzzy. However, FCM clustering image segmentation algorithm is time-consuming when dealing with large sample. In order to improve the processing speed, [6] introduces WFCM algorithm based on eigenvalue distribution of sample in the feature space. WFCM algorithm's computing performance is significantly better than FCM algorithm. In [7], it presents an image segmentation method on the basis of multiple gray-scale histogram, and it achieves fast segmentation according to the WFCM clustering algorithm. In [8], it puts forward a new image segmentation method of FCM clustering based on chroma information that is also the early work in this paper. For fast and accurate detection, optimized sample based WFCM clustering segmentation algorithm is presented.

The pitch area can be determined through regional growth method or boundary detection. Previous methods using progressive scanning to determine the pitch boundary as well as using least-squares fitting pitch boundary [9], are unable to eliminate the detection error when there is non-pitch-color objects appear in the pitch boundary. And Wang [10] calculates the convex shape of the two-dimensional points set to determine the pitch boundary by Sklansky algorithm. This method can meet the detection requirements basically. In our study, we use Hough transform to detect the pitch boundary.

The rest of the paper is organized as follows. WFCM clustering algorithm is presented in detail in Section 2. Hough transform is discussed in Section 3. Section 4 introduces the algorithm flow. Section 5 gives the experimental results and analysis. And Section 6 concludes the paper.

2. II. WFCM Clustering Segmentation

FCM, i.e. the fuzzy C-means clustering, is a typical image segmentation method using membership to determine the degree that each point belongs to a certain clustering.

978-0-7695-5050-3/13 $26.00 © 2013 IEEE
DOI 10.1109/ICIG.2013.65

2.1. The result of FCM segmentation

The process of FCM is choosing the proper fuzzy membership matrix and cluster center through a series of nonlinear iteration to minimize the target function. The FCM algorithm introduced membership matrix, in which the value of each data is within the interval [0, 1], and the sum of the data of the membership matrix is 1. Figure 1 shows the segmentation results after FCM clustering. From this example, it can be observed that FCM clustering algorithm can obtain a good segmentation result.

Figure 1: FCM clustering results

However, the general applications of FCM algorithm only consider gray-scale factors, so that the effect in most cases is not satisfactory. In this paper, we optimize the sample according to the characteristics of green pitch by weighting and summating the G-Y component in RGB space and the S component in HSV space, and the corresponding result is assigned to each pixel, i.e. each pixel value is (G-Y)×50%+S×50%. In addition, FCM algorithm is so time-consuming when dealing with a large sample that a histogram weighted FCM algorithm is used in our experiment.

2.2. WFCM clustering algorithm

WFCM is proposed on the basis of the concept of the weighted sample and the structure of the weighted set. Using the samples and sample weights to describe the sample eigenvalue distribution in different spaces, we can get the weighted sample.

WFCM algorithm takes the different clustering effects of different sample vector into consideration, so the weighted fuzzy c-means clustering of sample set X can be expressed as the following mathematical programming problems:

$$\min\{J_m(U,V)\} = \min\left\{\sum_{i=1}^{c}\sum_{k=1}^{n} w_k (u_{ik})^m (d_{ik})^2\right\} \quad (1)$$

where w_k represents the weighting coefficient for each sample set. Then use Lagrangian to derive two optimization iteration formulas (2) and (3):

$$u_{ik} = \left[\sum_{j=1}^{c}\left(d_{ik}/d_{jk}\right)^{2/(m-1)}\right]^{-1} \quad (2)$$

$$c_i = \sum_{k=1}^{n} w_k (u_{ik})^m x_k \Big/ \sum_{k=1}^{n} w_k (u_{ik})^m \quad (3)$$

$$w_k = n(k)/(M\times N) \qquad \sum_{k=1}^{L-1} w_k = 1, k = 0,1\ldots,L-1 \quad (4)$$

where n(k) indicates the occurrence times of the pixel with the value of k in the image, L denotes the total order of the pixel value.

Obviously, the weighting coefficients w_k is mainly used for the adjustment of the cluster centers. WFCM algorithm will degenerate to FCM algorithm if the weighting coefficients of each sample are the same. In WFCM algorithm, as long as w_k is selected properly, a more accurate clustering result will be obtained compared with the standard FCM algorithm.

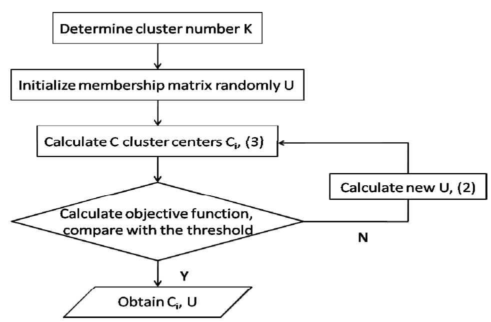

Figure 2: WFCM algorithm flowchart

The WFCM algorithm flow chart is shown in figure 2:

2.3. Comparison between WFCM and FCM

In section 2.1, the optimized sample is the weighting summation of G-Y and S. If the size of the image is 256×256, the number of samples will reach 65536, and the number of samples will grow as the image size increases, which greatly affect the real-time segmentation process. In view of this situation, we can use the frequency of pixel value as the sample to be classified instead of the pixel value. Therefore, the image size only affects the calculation of the histogram and even if the image size increases, the number of clustering samples will not change.

In order to compare WFCM with FCM, we use the two algorithms to process three images with different resolution respectively. The specific results are shown in table 1.

Table 1: algorithm performance comparison

Image size	The number of Iterations		Computation time (s)		Time saved
	WFCM	FCM	WFCM	FCM	
320×180	40.3	41	0.883	1.545	42.85%
400×301	26.3	29.3	0.812	3.114	73.92%
720×640	20	25.7	0.802	9.497	91.56%

In Table 1, time saved = (WFCM computation time-FCM computation time)/FCM computation time×100%. As shown in table 1, WFCM algorithm can greatly reduce the image processing time and the bigger is the image size, the higher is the time saved, which fully reflects the advantage of WFCM algorithm when dealing with large sample set.

3. Hough transform

Using Hough transform to determine the boundary of the pitch is feasible, since the pitch is a standard polygon and the boundary is straight. The pitch area has been basically separated in the binary image obtained from WFCM clustering segmentation. The inanition in the binary image can be fixed by erosion operation, after which the interference caused by the audience and the pitch line are basically removed, so that the boundary is obvious and easy to be detected through Hough transform.

Hough transform is barely used in boundary detection, because it's hard to sort out the right lines from all the detected lines. Therefore, we propose a screening method using the slope and the intersections which will be described in the next section.

4. Pitch detection algorithm

This section introduces the specific pitch detection algorithm. The flowchart is shown in figure 3.

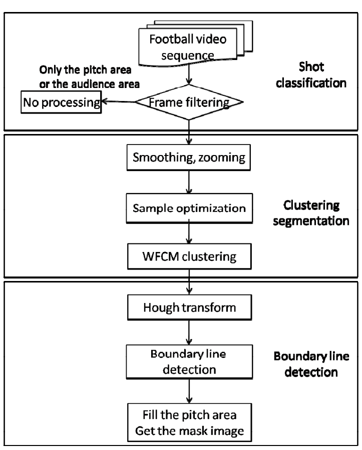

Figure 3: The pitch detection algorithm flowchart

(1) Shot classification. If the shot is filled with the audience area or the pitch area, as figure 4 shows, there is no need to do detection. In this step, pick up an 8×8 block in each corner of the image and select the frames need to be detected according to the statistical color feature.

Figure 4: Frames no need to be detected

(2) Clustering segmentation. Smooth and zoom the image in order to reduce the time for histogram calculation and pixel value classification. Weight and summate the G-Y and S components after the two components are shifted to the interval [0, 255] and then calculate the histogram, so that the weighting coefficient w_k could be obtained from the histogram values and the number of samples to be processed is 256, i.e. 0-255. Use WFCM clustering algorithm to process the image following the sequence in Figure 2.

(3) Boundary line detection. Figure 5 shows the flowchart in this part. Make morphological processing to the binary image. Find all of the lines in the binary image by Hough transform. Sort out the boundary lines of the pitch using slopes and intersections (see Figure 6). Once the boundary line is determined, the mask matrix of the pitch area could be obtained by filling the region inside the boundary.

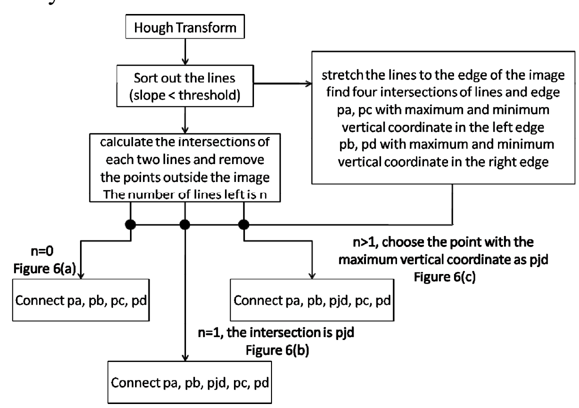

Figure 5: Boundary line detection flowchart

(a) (b) (c)

Figure 6: Three cases of the left intersections

5. Result and analysis

5.1. Compare different segmentation algorithms

Figure 7 shows the segmentation effects of several different algorithms.

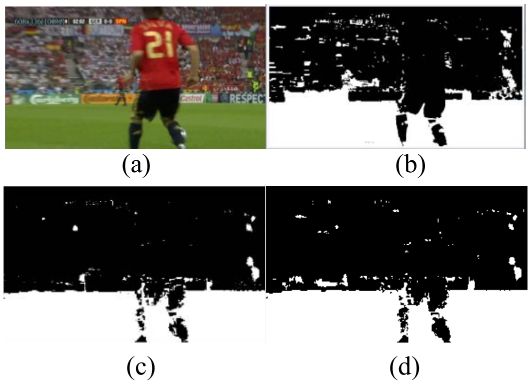

Figure 7: Different segmentation effects

Figure 7(a) is the original image. In Figure 7(b), a segmentation algorithm based on threshold is used. This is the simplest method which however is easy to be influenced by the light, shadow and many other factors. The result of K-means clustering is good in Figure 7(c) while the processing is time-consuming. WFCM clustering algorithm in this paper is on the basis of a fixed sample with a fast processing and a satisfactory result as is shown in Figure 7(d).

5.2. Compare different pitch boundary detection methods

Hough transform and Sklansky algorithms are used separately in experiments and the results are given in Figure 8.

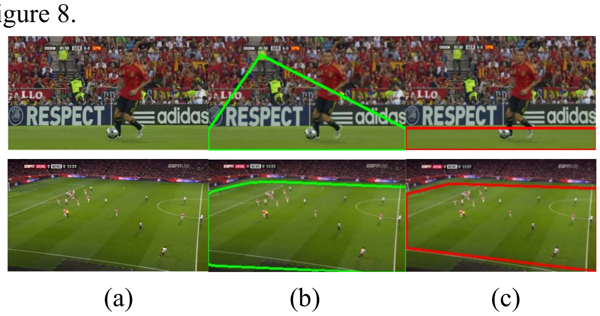

Figure 8: Different detection result

From Figure 8 we can see that, if the hole can't be removed by morphological operation, there will be a false detection using Sklansky algorithm.

5.3. Result analysis

To evaluate the proposed algorithm, we extracted 200 frames randomly from the Champion League, Premier League, La Liga and CSL. Some detection results of the four fragments are shown in Figure 9, wherein (a)(b)(c)(d) denote the original image, the binary image obtained from WFCM clustering segmentation, the results after Hough transform and the mask image, respectively.

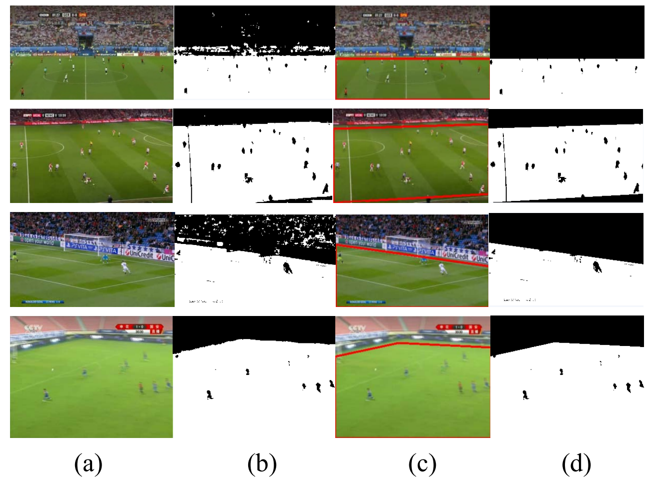

Figure 9: Some detection results

Table 2 lists the statistical results, including the time, the number of the frames need to be detected (NFND), the number of the frames in which the pitch area is detected correctly (NFCD) or wrongly (NFWD), and the number of frames undetected which should be detected (NFUD). From the table, we can find that the correct rate in each fragment is more than 90%, besides, the processing speed is fast.

Table 2: statistical results

Game fragment	Champion League	Premier League	La Liga	CSL
Image size	608×336	640×368	640×368	640×480
Time (s)	26	35	34	44
NFND	159	200	200	200
NFCD	149	185	191	181
NFWD	10	15	9	19
NFUD	0	0	0	0
Correct rate	93.70%	92.50%	95.50%	90.50%

6. Conclusions

This paper proposes a histogram weighted FCM clustering algorithm based on the eigenvalue of the color in two spaces. Moreover, a method that combines Hough transform and the shape characteristics of the pitch to determine the boundary line is presented. The proposed detection method is mainly used for football videos. In the future research, the method could be improved to apply in stadium detection in different kinds of match, such as tennis, badminton etc or alternatively to detect the players, the ball, the pitch line and so on.

Acknowledgment

This work is supported by the National Science & Technology Pillar Program of China under Grant No.2012BAH39F02.

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